CARL-DX — KOTI BUDGET SURVEY & PRICE COMPARISON WORKSHEET

PHYSICAL EMBODIMENT PLANNING • 100% SOLDERLESS (PIN & SOCKET / SCREW CLAMP)

📝 MARKET SURVEY / BUDGET PLANNING

⚡ ZERO-SOLDER

TARGET CEILING: ₹2,500 | Sim-to-Real: `vessel_kinetic.xml`

PURPOSE OF VISIT: Market price estimation, verifying solderless stock availability, and collecting quotations before purchasing.

Target Area: Gujarati Galli & Troop Bazaar, Koti

🗣️ **NO-PRESSURE SURVEY SCRIPT (How to ask for prices without feeling obligated to buy):**

- *"Bhaiya, M.Tech/B.Tech robotics project submission ke liye **budget estimate aur rough quotation slip** bana rahe hain. In components ka rate bata sakte ho kya?"*
- *"Agar yeh poore 15-20 items ek saath lenge (complete student kit pack), toh total bill pe kitna **package discount** milega?"*
- *"Bhaiya, mujhe soldering nahi aati, kya aapke paas yeh saare boards **pre-soldered pins** ke saath milenge, ya counter pe solder karke doge?"*

1. Core Controller & Motor Driver (PNS Subsystem)

Target Range: ₹420 – ₹490

#	COMPONENT	SOLDERLESS SPEC & SURVEY INQUIRY	QTY	TARGET	SHOP 1 (₹)	SHOP 2 (₹)	SOLDERED?
1	**ESP32 Dev Module** ESP-WROOM-32 (30-Pin) ✓ PRE-SOLDERED PINS	Dual-core 240MHz, Wi-Fi+BLE, micro-USB port (CP2102/CH340). 👁 Ask: "30-pin narrow board hai na (male pins already soldered downward)?" *Fallback: 38-Pin (₹290) or ESP32-CAM + OV2640 (₹420).*	**1**	₹300–330			[ ] Yes [ ] No
2	**Motor Driver Module** DRV8833 or L298N Mini ✓ PINS / SCREW TERMINAL	**Option A:** DRV8833 MOSFET (ask if male header pins are pre-soldered). **Option B:** L298N Mini / Standard L298N (has built-in screw terminals for motor wires). 👁 Ask: "Screw terminal wala driver hai kya ya pins soldered hain?"	**1**	₹55–80			[ ] Yes [ ] No
3	**Micro-USB Data Cable** 4-Wire Sync & Flash Cable ✓ PLUG & PLAY USB	Connects ESP32 to PC. Must support data transmission for programming. 👁 Ask: "Data sync cable hai na (sirf charging wali nahi honi chahiye)?"	**1**	₹40–60			N/A (Cable)

2. Spatial Perception & Inertial Sensors (mj_ray & CANN Grounding)

Target Range: ₹650 – ₹740

#	COMPONENT	SOLDERLESS SPEC & SURVEY INQUIRY	QTY	TARGET	SHOP 1 (₹)	SHOP 2 (₹)	SOLDERED?
4	**VL53L0X Laser ToF** Time-of-Flight Ranging (I2C) ✓ PRE-SOLDERED PINS	STMicro 940nm laser (30–2000mm). Physical `mj_ray`: Front, Left, Right. 👁 Ask: "VL53L0X pe pins pehle se soldered hain kya? Per-piece rate kitna hai?" *Fallback: VL53L1X (₹180/ea) or HC-SR04 Ultrasonic (₹65/ea).*	**3** (4 opt)	₹450–490 (~₹150/ea)			[ ] Yes [ ] No
5	**TCA9548A I2C Mux** 8-Channel I2C Bus Switch ✓ PRE-SOLDERED PINS	Connects multiple VL53L0X sensors sharing address `0x29`. 👁 Ask: "TCA9548A board with soldered header pins available hai?"	**1**	₹75–90			[ ] Yes [ ] No
6	**GY-521 MPU-6050 IMU** 6-DOF Accel + Gyroscope ✓ PRE-SOLDERED PINS	Measures yaw rate & linear slip acceleration. Direct I2C socket link. 👁 Ask: "GY-521 module blue color board with header pins rate kya hai?"	**1**	₹120–140			[ ] Yes [ ] No

🔧 Counter Soldering Service Inquiry (If boards have loose pins)

Ask the counter in-charge or technician: "*Bhaiya, agar yeh 5 boards (ESP32, 3x ToF, Mux, IMU) ke loose pins yahi counter pe solder karwa ke lenge, toh kitna service charge loge?*" (Standard market charge in Gujarati Galli is **₹10 – ₹20 total** for all boards).

CARL-DX — POWER, CHASSIS & SOLDERLESS CONNECTORS

PRICE SURVEY & STOCK INQUIRY FOR HARDWARE, BATTERY & WIRING

PAGE 2 SURVEY

SCREW & SPRING CLAMPS

Target Subtotal: ₹850 – ₹1,020

3. Autonomous Power & Solderless Voltage Regulation

Target Range: ₹360 – ₹420

#	COMPONENT	SOLDERLESS SPEC & SURVEY INQUIRY	QTY	TARGET	SHOP 1 (₹)	SHOP 2 (₹)	STOCK TYPE
7	**18650 Li-ion Cells** 3.7V Rechargeable (2000–2600mAh) ✓ STANDARD SNAP-IN	Two cells in series = 7.4V nominal. Snaps into holder with zero tools. 👁 Ask: "Original 2000-2600mAh brand kaunsa hai? Multimeter pe voltage check karoge?" *Note: Reject suspiciously cheap cells claiming "9900mAh" or weighing <35g!*	**2**	₹180–220 (~₹100/ea)			Brand: ____
8	**2S 18650 Battery Case** Dual Series Box with Leads ✓ WIRE LEADS	Holds two 18650 cells in series (7.4V). 👁 Ask: "2S dual battery holder with thick wire leads hai kya?"	**1**	₹30–45			[ ] With Leads
9	**LM2596 Buck Module** Step-Down with Screw Terminals ✓ SCREW CLAMP BLOCKS	Steps 7.4V battery down to clean 5.0V for ESP32 VIN. 👁 Ask: "LM2596 buck screw terminal block wala hai kya (jisme wires screw se tight hoti hain)?" *Fallback: Mini360 Buck (ask: "pins solder karke doge?").*	**1**	₹50–65			[ ] Screw [ ] Pin
10	**CH-2 Quick Connectors** Spring-Lever 2-Pin Blocks ✓ PRESS-TO-CLAMP	WAGO-style toolless wire connectors. Finger press → insert wire → release clamp! 👁 Ask: "CH-2 ya press-type 2-pin terminal block connectors milenge?"	**4 pcs**	₹20–30 (~₹5/ea)			[ ] Available
11	**2S 8.4V Li-ion Charger** Wall Adapter or TP5100 Module ✓ PLUG-IN DC JACK	Recharges 7.4V pack safely. Ask rate for 8.4V 1A DC adapter vs TP5100 module.	**1**	₹60–90			[ ] Adapter [ ] Mod

4. Mobile Chassis & Locomotion Hardware

Target Range: ₹350 – ₹420

#	COMPONENT	SOLDERLESS SPEC & SURVEY INQUIRY	QTY	TARGET	SHOP 1 (₹)	SHOP 2 (₹)	STOCK TYPE
12	**2WD Smart Chassis Kit** Acrylic Deck + 2 BO Motors + Wheels ✓ BOLT-TOGETHER ASSEMBLY	Acrylic plate, 2x 1:48 yellow BO motors, 2x rubber wheels (66mm), caster, screws. 👁 Ask: "Motors pe lead wires pehle se lage hain ya alag se solder karwa ke doge?"	**1 kit**	₹260–300			[ ] Wires on [ ] Bare
13	**CY-15A Metal Ball Caster** Cushioned Caster / Omni Ball Unit ✓ SCREW MOUNT	Metal ball transfer unit to eliminate chassis pitch teeter-totter wobble. 👁 Ask: "CY-15A ball caster ya standard metal ball caster hai kya?"	**1**	₹50–75			[ ] Available

5. Solderless Interconnects, Breadboard & Tools

Target Range: ₹140 – ₹180

#	COMPONENT	SOLDERLESS SPEC & SURVEY INQUIRY	QTY	TARGET	SHOP 1 (₹)	SHOP 2 (₹)	STOCK TYPE
14	**Dupont Jumper Wires** 40-Pin Ribbon Wires (20cm) ✓ PIN & SOCKET JUMPERS	Need 3 packs: **1x F-F** (sensors to mux), **1x M-F** (boards to breadboard), **1x M-M** (rails). 👁 Ask: "Teeno types (F-F, M-F, M-M) ke 40-wire packets ka bundle rate kya hai?"	**3 packs**	₹90–110 (~₹35/pkt)			[ ] All 3 types
15	**400-Point Breadboard** Half-Size Solderless Breadboard ✓ 100% SOLDERLESS PUSH-FIT	Self-adhesive back mounts directly on robot deck for 3.3V/GND socket rails.	**1**	₹45–60			[ ] Available
16	**Mini Screwdriver** Dual Flathead / Phillips (2mm) ✓ FOR SCREW TERMINALS	Tightens screw terminals on LM2596 buck converter and M3 chassis bolts.	**1**	₹15–25			[ ] Available

TOTAL ESTIMATED OUT-OF-POCKET EXPENSE (16 ITEMS)	₹1,920–2,250	₹ _____	₹ _____	≤ ₹2,500 Goal

CARL-DX — MARKET SURVEY GUIDE & BUDGET TIERS

SHOP DIRECTORY, ESTIMATION SLIPS & 3 FORMAL BUDGET TIERS

📍 PAGE 3 PLANNING GUIDE

3 BUDGET TIERS DEFINED

📍 Koti Component Shops to Visit for Quotes

- **Gujarati Galli (Main electronics lane):**
  - **Universal Electronics / Universal Components:** Best for ESP32, ToF sensors, and pre-soldered module breakouts. Ask for Mr. Ramesh/counter manager.
  - **Sagar Electronics / Sagar Components:** Excellent competitive pricing on sensors, buck converters, and multiplexers.
  - **Metro Electronics / Shweta Electronics:** Great for 18650 cells, battery cases, chargers, breadboards, and jumper ribbons.
- **Troop Bazaar (Hardware & Motors):**
  - **Vishal Electronics & Pioneer Electronics:** Best for 2WD chassis kits, BO gearmotors, rubber wheels, casters, brass spacers & screws.
- **How to Walk the Market:** Start at Gujarati Galli (Bank Street entrance) → survey electronics/sensors → walk through to Troop Bazaar for chassis & motors.

📝 How to Request an "Estimation Slip"

- **College Requirement Pretext:** Always say: "*Bhaiya, guide/college approval ke liye quotation estimate chahiye, sanction aane pe poora set yahi se khareedenge.*"
- **They Will Write it on a Memo:** Most shops will write down individual rates on their small printed rough memo slip with their shop stamp/visiting card!
- **Collect Visiting Cards:** Take the shop's visiting card and staple/attach their rough estimate slip so you can compare prices at home tonight.
- **Ask About Package Discount:** Always ask: "*Agar 16 items ka complete package lein, toh kitna kam karoge?*" (Usually 5% to 10% lower than individual items).
- **Take Multimeter for Reference:** If you have a small multimeter, keep it in your bag to verify batteries when you return to buy.

CARL Physical Embodiment — 3 Formal Budget Tiers

Dissertation Budget Scenarios

Tier A: Bare-Bones Minimal Frugal — Est: ₹1,850 – ₹2,050

Core differential drive + 2x ToF sensors (Front & Flank) + ESP32 + 18650 pack + Drag caster.

- Sufficient for Stage 1 Body Babbling & Stage 2 2D raster memory exploration.
- Eliminates I2C multiplexer (uses direct GPIO XSHUT pins for ToF addressing).
- Standard plastic drag caster included in ₹280 chassis kit.

Tier B: Recommended Complete Solderless — Est: ₹2,150 – ₹2,400

Full 3x ToF array (0°, ±90°) + TCA9548A mux + MPU-6050 + CY-15A suspension caster + CH-2 lever blocks.

- 1:1 exact mathematical match with `vessel_kinetic.xml` in MuJoCo!
- 100% solderless plug-and-play assembly (spring clamps + Dupont).
- Smooth caster suspension eliminates drive wheel lift-off & pitch wobble.

Tier C: Optional Active Vision & Telemetry Upgrades (Stage 3.7 Add-Ons)

Optional Add: ~₹250 – ₹550

#	COMPONENT	PLUG-AND-PLAY ROLE / SPECIFICATION	QTY	TARGET	SHOP 1 (₹)	SHOP 2 (₹)	NOTES
17	**ESP32-CAM + OV2640** Camera Dev Board	Adds 64x64 retinal vision stream for Stage 3.7 gaze. Ribbon camera snaps into connector.	1	₹380–450			Vision stream
18	**INA219 Current Sensor** I2C Power Monitor	Pre-soldered screw terminals! Measures real-time motor load for `carl_fatigue.py`.	1	₹110–130			Torque/Load
19	**SG90 Micro Servo (9g)** Pan Actuator (3-pin plug)	Standard 3-pin servo plug. Controls active pan head gaze (±0.85 rad). Zero solder.	1	₹70–85			Pan gaze

📝 Market Survey Notes & Shop Visiting Cards: